Task 1: 3 Features of Zomato Improved by Fine-Tuning

• Hyper-Local Review Summarization: Fine-tuning an AI model on Indian culinary vocabulary allows it to extract highly specific sentiments from customer reviews (e.g., distinguishing between authentic "Kathiyawadi spice" and accidental over-spicing) rather than just broad positive or negative ratings.

• Contextual Search and Discovery: Fine-tuning helps the search bar understand regional slang, abbreviations, or misspelled dish names (like "Gujju thali", "sev khamani", or "dhokla") to immediately map them to the correct restaurant menus.

• Automated Customer Support Escalation: Fine-tuning on past customer support logs trains the chatbot to recognize complex delivery disputes, calculate refund eligibility based on company policy, and draft empathetic, accurate resolutions without human intervention.

Task 2: Model Comparison for Netflix Movie Recommendations

Feature / Metric	Pre-trained AI Model	Fine-tuned AI Model
Accuracy	Provides broad, generic accuracy; recommends global box-office hits but misses individual user preferences.	High precision accuracy; learns exact historical viewing habits, watch times, and skipped titles of specific user profiles.
Relevance	Suggests trending content that often mismatches a user's current mood, language preference, or niche taste.	Highly relevant suggestions that adapt dynamically to subtle sub-genres, favorite directors, and regional content preferences.
User Experience	Leads to choice fatigue, forcing users to spend significant time scrolling through irrelevant titles to find a movie.	Highly engaging and satisfying; presents appealing choices directly on the home banner, increasing immediate watch-time.

Task 3: Chatbot Scenario (Swiggy Order Help)

Scenario: A customer contacts support because their food order arrived damaged and missing items, requiring the bot to verify uploaded photos against internal refund rules and historical account validity.

Reasoning: Prompt engineering alone cannot safely handle this because it relies on static text instructions that are easily bypassed by creative user complaints or variations in language. It also cannot safely memorize complex, frequently updated internal company refund structures and customer account states without risking data hallucination. Fine-tuning is required to embed strict adherence to company compliance guidelines directly into the model's behavioral weights, ensuring reliable, legally sound decisions across thousands of unique edge cases.

Task 4: Recommendation System Strategy for Gujarat Regional Music

Decision: The company should use Fine-Tuning.

• Reason 1 (Domain-Specific Knowledge): Fine-tuning embeds deep cultural familiarity with specific Gujarati musical traditions (such as Garba, Dayro, Sugam Sangeet, and Bhajans) directly into the model, allowing it to correctly categorize and relate songs that standard pre-trained models do not understand.

• Reason 2 (User Personalization): It allows the system to recognize regional user behavioral patterns—such as the massive shift toward folk and dance music during the Navratri festival season—optimizing playlist recommendations based on cultural context rather than just generic listening patterns.

Task 5: Food Delivery Chatbot Prompt & Extension

Example Prompt

"You are an expert culinary assistant specializing in Gujarati cuisine. When a user asks for food recommendations, respond enthusiastically and include warm Gujarati hospitality phrases (such as 'Aavjo' or 'Jamva Padharo'). Suggest authentic dishes like Locho, Undhiyu,

Handvo, or Shrikhand depending on whether they want a light snack or a full meal, and always maintain a polite, food-loving persona."

How Fine-Tuning Enhances the System

Fine-tuning enables the model to accurately recognize and cross-reference hundreds of local restaurant menus, regional dialect variations, and custom ingredients across Gujarat without exceeding prompt token limits. It also ensures the chatbot maintains its specialized cultural persona consistently during long, complex multi-turn conversations without experiencing hallucination.